

Topic

An Intelligent Search and Rescue System to Detect Lost Humans in The Desert

Abstract

The system integrates various components, including thermal imaging sensors, and an Artificial intelligence (AI) algorithm for data processing and decision-making. airborne patrol predefined search areas, capturing real-time imagery and thermal data. This data is transmitted to a central command center where the AI algorithm processes it to identify potential human targets based on thermal signatures, body shapes, and movement patterns.

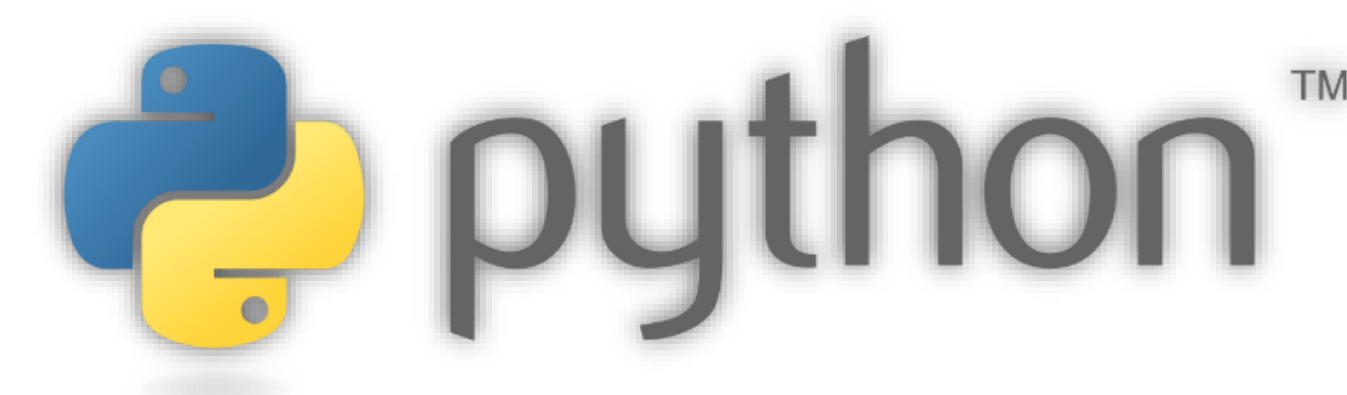
The proposed system offers several advantages over traditional search and rescue methods. It reduces reliance on manpower and costly resources by employing airborne equipped with advanced sensors. Additionally, the integration of AI enhances the efficiency and effectiveness of search operations by rapidly analyzing vast amounts of data to identify potential targets.

Objectives

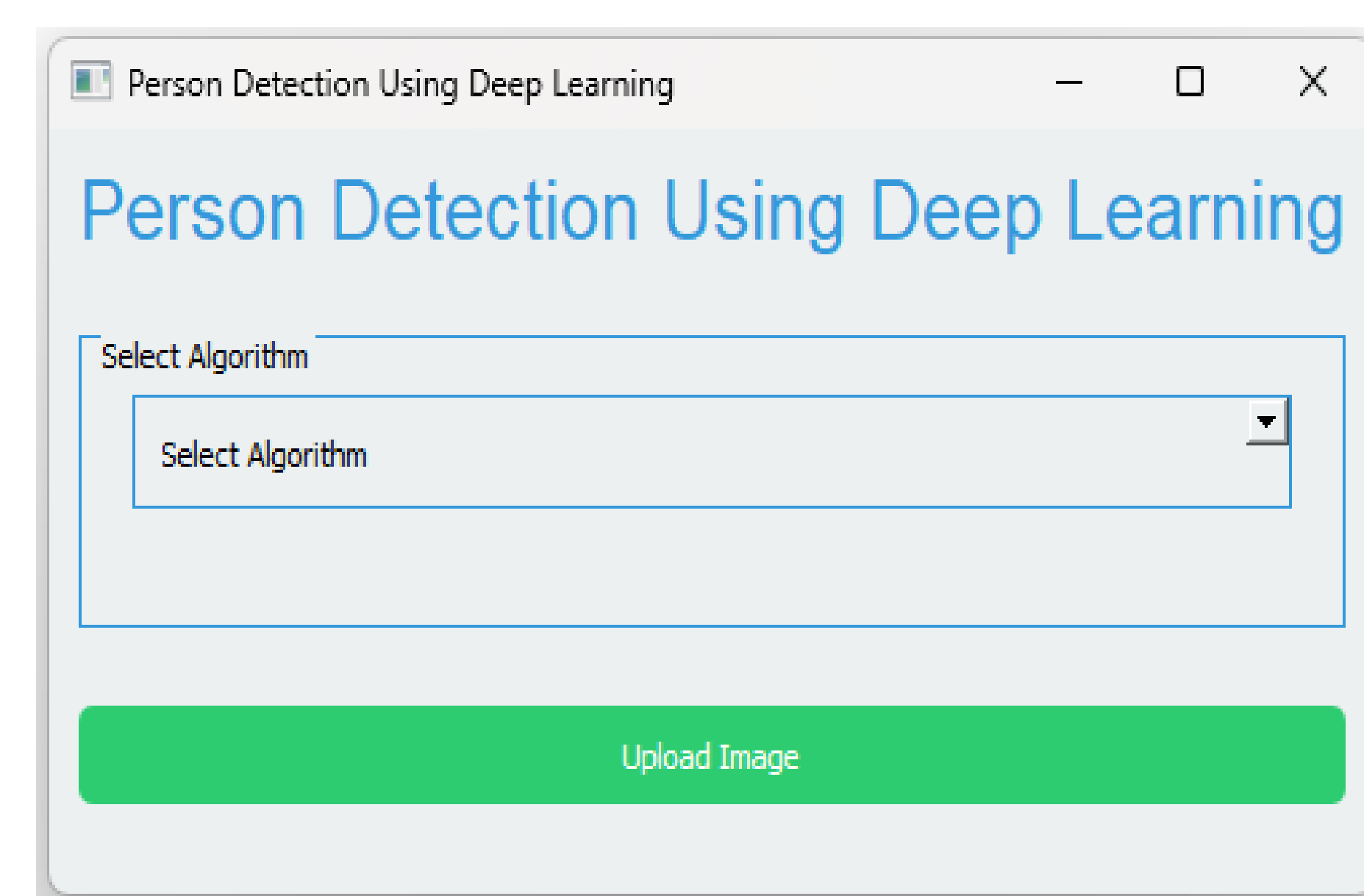
- ❑ To investigate similar systems related to search and rescue.
- ❑ To design an algorithm for detecting lost humans by Image processing technique.
- ❑ To develop different algorithms to find lost people.

project tools

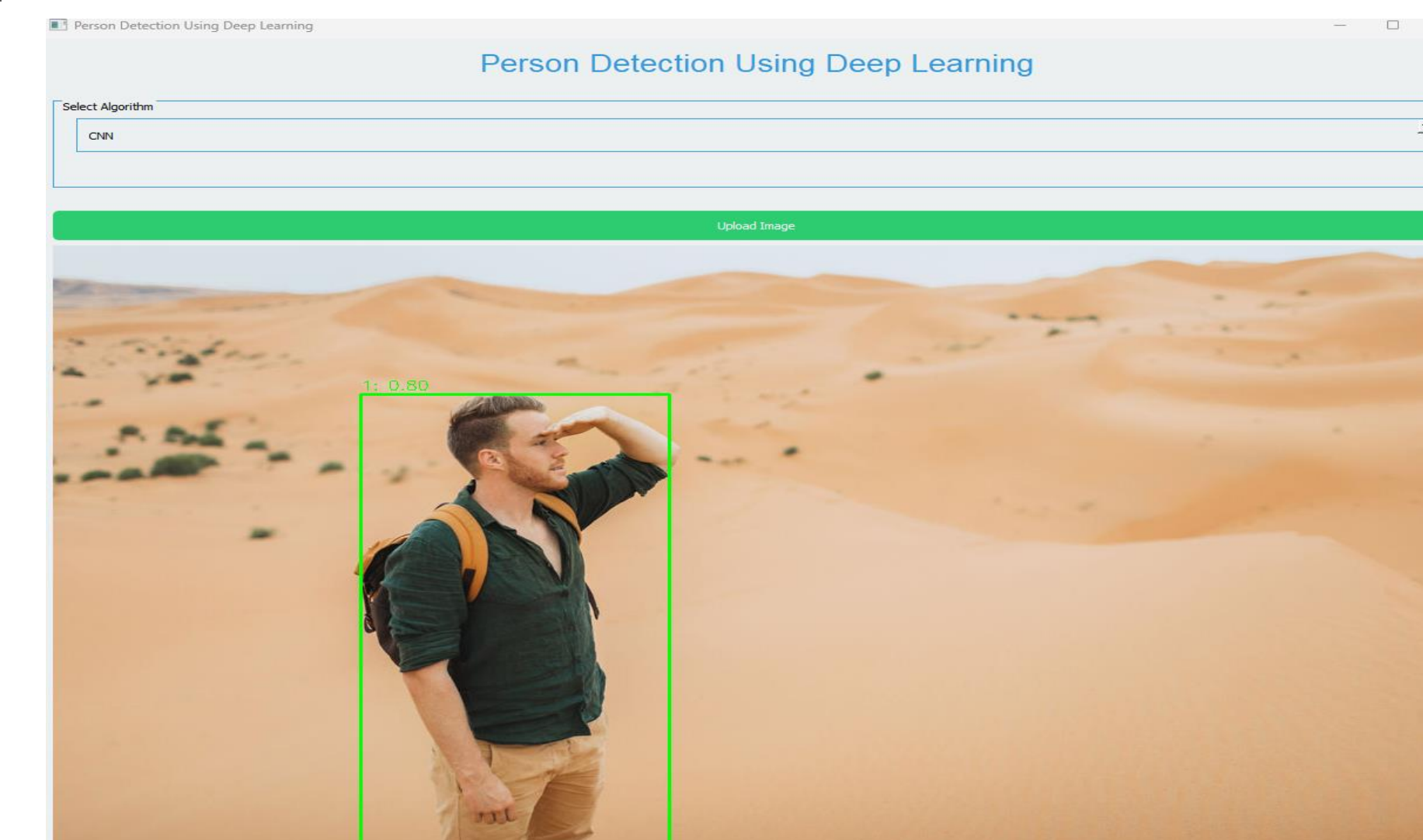
The project is implemented by python language. Python is one of effective language which used in different application such as deep learning and machine learning.



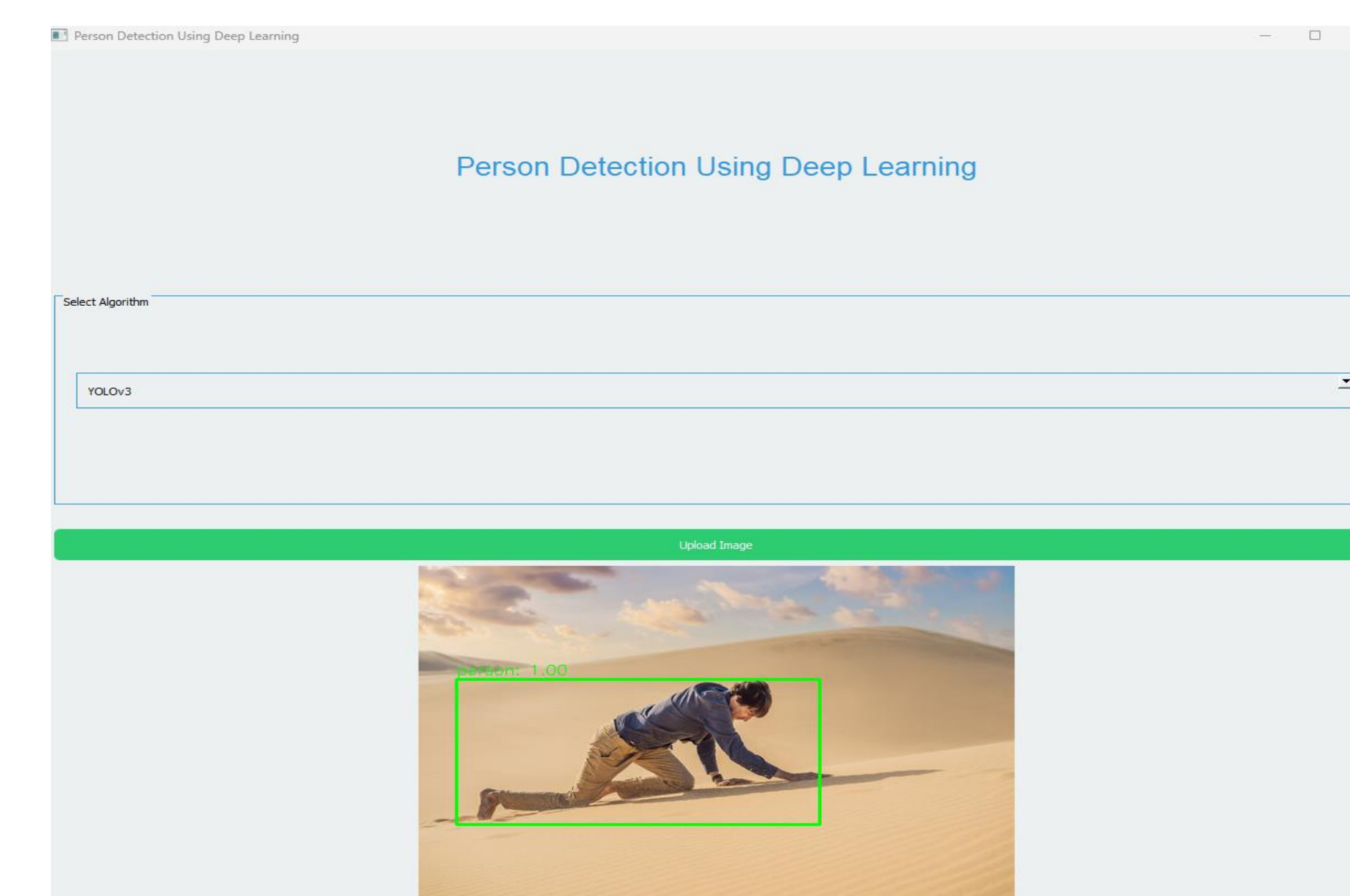
Project interface



Project interface/ CNN



Project interface/ YOLO

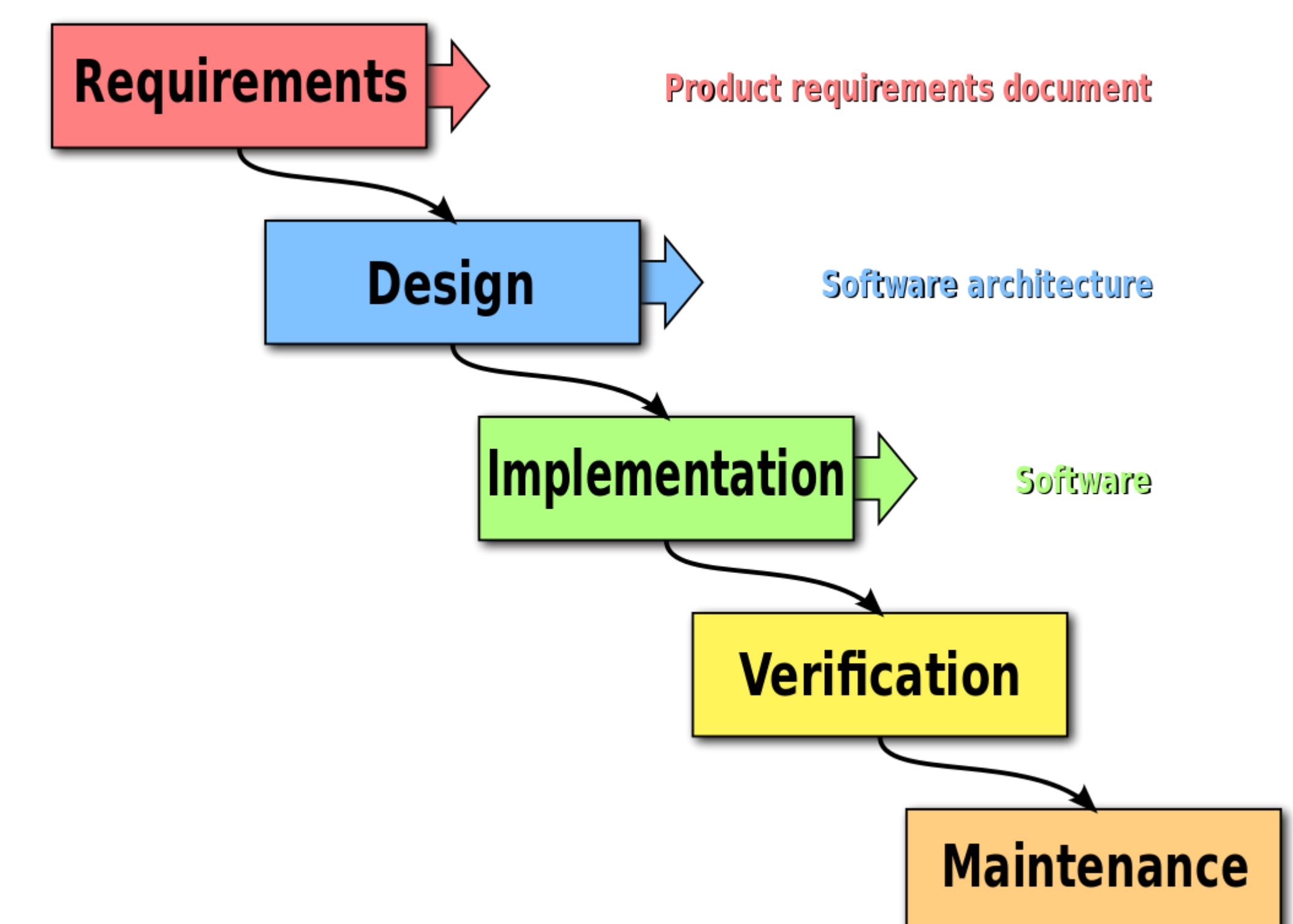


COCO database

'person', 'bicycle', 'car', 'motorcycle', 'airplane', 'bus', 'train', 'truck', 'boat', 'traffic light', 'fire hydrant', 'stop sign', 'parking meter', 'bench', 'bird', 'cat', 'dog', 'horse', 'sheep', 'cow', 'elephant', 'bear', 'zebra', 'giraffe', 'backpack', 'umbrella', 'handbag', 'tie', 'suitcase', 'frisbee', 'skis', 'snowboard', 'sports ball', 'kite', 'baseball bat', 'baseball glove', 'skateboard', 'surfboard', 'tennis racket', 'bottle', 'wine glass', 'cup', 'fork', 'knife', 'spoon', 'bowl', 'banana', 'apple', 'sandwich', 'orange', 'broccoli', 'carrot', 'hot dog', 'pizza', 'donut', 'cake', 'chair', 'couch', 'potted plant', 'bed', 'dining table', 'toilet', 'tv', 'laptop', 'mouse', 'remote', 'keyboard', 'cell phone', 'microwave', 'oven', 'toaster', 'sink', 'refrigerator', 'book', 'clock', 'vase', 'scissors', 'teddy bear', 'hair drier', 'toothbrush'

Project Model

We use prototyping model in our project, This model works best in scenarios where not all the project requirements are known in detail ahead of time. It is an iterative, trial-and-error process that takes place between the developers and the users..



conclusion

In conclusion, leveraging artificial intelligence (AI) and drones for search and rescue operations in desert environments presents a promising solution to the critical challenge of locating lost individuals efficiently and effectively.

We identify the project design, so we have clear information about system design and system architecture